fpga_cpu: Spartan-3 Starter Board expansion (rough). Serial and the keyboard are on the board itself (DB9: RXD T13, TXD R13; PS/2: M15/M16).

#FLG01 PWR_FLAG
#FLG02 PWR_FLAG
#FLG03 PWR_FLAG

A2 expansion connector (UG130 table 13-3)
4 E6 I2S_BCK 5 D5 I2S_LRCK 6 C5 I2S_DATA
7 D6 MS_CLK 8 C6 MS_DATA
9 E7 SD_CS 10 C7 SD_SCK 11 D7 SD_MOSI
12 C8 SD_MISO 13 D8 SD_CD
14 C9 PAD_SEL (out) 15 D10 PAD1_UP 16 A3 PAD1_DOWN 17 B4 PAD1_LEFT
27 B10 PAD1_RIGHT 28 A10 PAD1_TL 29 B11 PAD1_TR 30 B12 PAD2_UP
31 A12 PAD2_DOWN 32 B13 PAD2_LEFT 33 A13 PAD2_RIGHT 34 B14 PAD2_TL
35 D9 PAD2_TR (the oscillator socket's pin: leave the socket empty)
18 A4 WIFI_CS 19 B5 WIFI_SCK 20 A5 WIFI_MOSI 21 B6 WIFI_MISO
22 B7 WIFI_HS 23 A7 WIFI_EN 24 B8 WIFI_BOOT 25 A8 WIFI_TXD 26 A9 WIFI_RXD
36-40 configuration: leave open
2 +5V (VU), 3 +3.3V (VCCO), 1 GND

Sound: I2S (64 BCK a frame, 48,828 Hz) into a PCM5102A: SCK tied low; the DAC's PLL makes its clock from BCK. FMT low = I2S. XSMT high = unmuted. Line out through a 470R / 2.2nF filter.

PS/2 mouse (the keyboard uses the board's own PS/2 port, M15/M16). Open drain both ways: the FPGA drives low or lets go. BSS138 level shifters (gate at 3.3 V); 10k pull-ups on both sides. Not 5 V tolerant!

WiFi: ESP32-WROOM-32E (e.g. ESP-A1 firmware) on its own pins, not the SD bus: shared, the ESP's MISO would be GPIO12, a strap the SD pull-up would hold high (1.8 V flash; no boot). VSPI slave: SCK IO18, MOSI IO23, MISO IO19, CS IO5; IO4 = handshake/interrupt to the FPGA; EN (reset) and IO0 (download mode) driven by the FPGA (open drain: pull-ups, EN RC). UART1 (IO16 in, IO17 out) to the FPGA too: UART0 on J8 for flashing. Own AMS1117 from +5V: the module draws ~500 mA when transmitting. One SPI engine in the FPGA serves SD and WiFi.

microSD in SPI mode: DAT3 = CS, CMD = MOSI, DAT0 = MISO; DAT1/DAT2 pulled up (unused); card detect to the FPGA, 47k pull-ups, 33R on the clock.

A1 to the Spartan-3 board)

A1 expansion connector (UG130 table 13-2): shared with the board's SRAM
5-19 odd: IC10 IO0-IO7 (sram_d[7:0]): A0-A17: pins 6, 8, 10, 12, 14, 24, 23, 26, 25, 28, 27, 30, 29, 32, 31, 34, 33, 35; 16 WE# 18 OE#
21 M7 ROM_CE (unshared), 20 P9 ROM_A18 (DOUT/BUSY: toggles while the FPGA configures, harmless on an address line); free: 4 N8, 22 M10
36-40: JTAG (TMS, TCK, TDO): do not connect

NOR flash ROM, executed in place (a resident toolbox: no RAM spent on it). The part: SST39VF1601C-70 (1M x 16, 3.3 V, TSOP-48), DQ0-DQ7 used: 512 KB. Its A0-A18 as shown, A19 to GND, DQ8-DQ15 open (pulled up in the FPGA's simulation only). WP#/RST# to 3.3 V. (The symbol is a stand-in: KiCad has none for it; the connections are the same by name.) CE# pulled up: off while the FPGA configures. Writes (programming) use the JEDEC command sequences through the same bus, with a long WE# pulse.

J3 needs a mini-DIN-6 footprint (e.g. CUI MD-60S): none in KiCad's library.

Sega Genesis / Mega Drive pads (3- and 6-button) on DE9 plugs, at 5 V (each through a 500 mA polyfuse). One select (TH) for both pads, driven at 5 V by U7; 100R to each port. The 12 inputs come back through two 74LVC245A at 3.3 V (5 V-tolerant inputs); the Spartan-3's pins are not. TH high: UP DOWN LEFT RIGHT B C; TH low: UP DOWN 0 0 A Start. A 6-button pad answers its 3rd low with UP..RIGHT all 0, and its 4th high with Z Y X Mode; it resets 1.5 ms after the last pulse. Software reads both pads. With TH low, LEFT and RIGHT at 1 (the pull-ups): no pad.

rough: sound, PS/2 mouse, microSD, Genesis pads, ESP32 WiFi (A2); NOR flash ROM on the SRAM bus	
Sheet: / File: s3_expansion.kicad_sch	
Title: fpga_cpu Spartan-3 expansion	
Size: A2	Date:
KiCad E.D.A. 10.0.5	Id: 1/1
	Rev: 0.1